

Highly Efficient Dual Wavelength Pumping Scheme for Thulium-Doped Fiber Amplifier

Mike Nix and Scott S-H. Yam, *Member IEEE*

Department of Electrical and Computer Engineering, Queen's University
Walter Light Hall, Kingston, ON K7L 3N6, Canada
Email: 3mn11@qlink.queensu.ca

Abstract—A novel 690 and 1400 nm dual wavelength pumping scheme for TDFA is proposed. Additional gain improvement of 6 dB is achieved at similar pumping level as previous demonstrations.

OCIS codes: (060.2320) Fiber optics amplifiers and oscillators; (140.4480) Optical amplifiers; (160.5690) Rare earth doped materials.

I. INTRODUCTION

Coarse wavelength-division multiplexing (CWDM) systems allow larger wavelength spacing between channels, thus relaxing the temperature and wavelength sensitivities of network components, and their subsequent costs. Thulium-doped fiber amplifier (TDFA) is a potential enabling component in such systems as it opens up the low-loss S-band window in optical fibers, increasing the total bandwidth available for CWDM. Many pumping schemes have been proposed to provide successful signal amplification [1]–[8] in TDFA, and it is imperative that high gain is achieved efficiently in terms of both pump power and component cost.

In the paper by [1], an experimental study of a TDFA with a 690 nm forward pump and 1400 nm backward pump was carried out. In this study, a numerical model was created to encapsulate these experimental results and was then used to investigate in detail TDFA gain characteristics. A novel pumping scheme is proposed which provides substantial improvement in gain given the same pumping levels as in the previous study.

II. MODEL CONFIGURATION

Fig. 1 shows the energy level diagram for Tm^{3+} and the transitions taken advantage of by the 690 nm and 1400 nm scheme. The 690 nm pump populates the $^3\text{F}_4$ level through non-radiative decay from the $^3\text{F}_3$ level, where the 1400 nm pump helps maintain inversion by excited-state absorption (ESA) from $^3\text{H}_4$ to $^3\text{F}_4$, while also populating $^3\text{H}_4$ from the ground state.

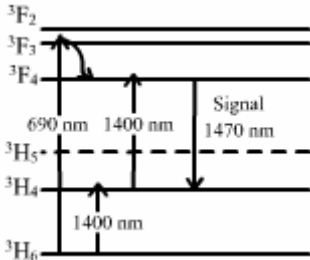


Fig. 1. Energy level diagram for proposed pumping scheme.

Fig. 2 shows the proposed setup of this TDFA. The fiber is divided into two sections, the first being nine meters long and the second four meters. These lengths were chosen because model simulations showed that they allowed for the most gain in shortest total amplifier length. The 1400 nm pump is sent backwards through both sections at a power of 79 mW while the 690 nm pump is sent forward at the start of each section at 29 mW.

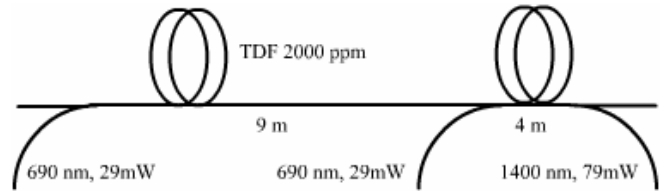


Fig. 2. Proposed TDFA configuration.

The model itself was created using parameters such as absorption and emission cross-sections and decay rates found in published literature [2], [9]–[13]. Because the model was created for Tm in ZBLAN glass (unlike in published literature), some of the decay rates had to be modified so that the model more accurately reflected experimental data.

III. RESULTS AND DISCUSSION

Fig. 3 and 4 show the results of the experiment conducted by [1] compared to the calculated results generated by the model. In Fig. 3, the 690 nm pump power was set to 58 mW and the 1400 nm pump power to 79 mW while gain was measured and calculated over a range of wavelengths from 1468 to 1508 nm with an input power of -18 dBm per channel. In Fig. 4, the 690 nm pump power was varied from 10 to 58 mW and the 1400 nm pump power was varied from 0 to 79 mW while gain was measured and calculated at a wavelength of 1482 nm and input power of -35 dBm. The gain results from the model match up quite well with the results from experiment, justifying the use of the model in simulations with similar operating conditions, such as with the proposed pumping scheme.

Fig. 5 shows the gain results generated by the model for the new pumping scheme. The total pump powers for this scheme remained the same (58 mW total 690 nm power, 79 mW total 1400 nm power); however, gain is increased over previous configurations by as much as 6 dB. Using this pumping scheme, more than 12 dB of gain over 40 nm in the S-band is achievable, with a maximum gain of almost 25 dB.

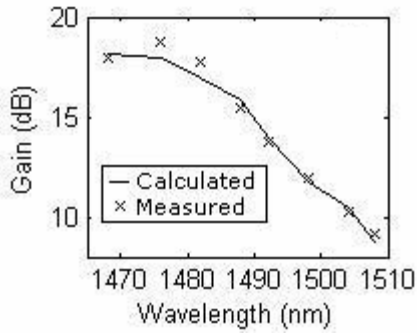


Fig. 3. Comparison of calculated and measured gain for multiple wavelengths.

In this paper only the gain characteristics of this novel amplifier configuration have been studied, while noise figure will be the focus of future work.

Since the pump powers used to achieve relatively high gain are quite low, this scheme is very efficient and shows great promise for TDFAs in future communication networks. Also, because digital video disk (DVD) systems use laser diodes (LD) of wavelengths close to 690 nm, a readily available and inexpensive DVD LD can be used to pump this system [14].

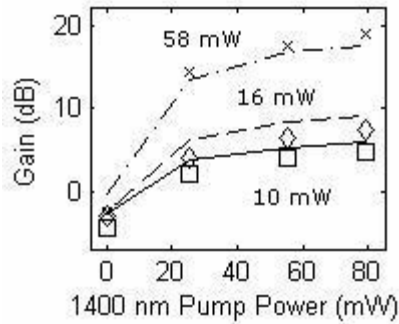


Fig. 4. Comparison of calculated (lines) and measured gain for varying pump powers. 690 nm pump power increases upward.

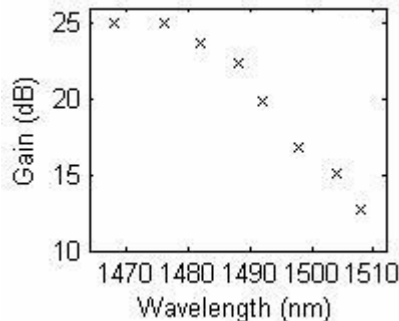


Fig. 5. Gain results from numerical model for proposed pump scheme.

IV. CONCLUSIONS

In this paper, a novel pumping configuration for TDFA is proposed, showing more than 12 dB of gain over forty nanometers in the S-band, with a maximum gain of almost 25 dB. Gain improvement of 6dB is obtained at low pump power levels similar to previous demonstrations. This scheme also uses a primary pump that can be generated by readily available DVD LDs, allowing for a relatively inexpensive implementation. Because of the possibility of highly efficient pumping schemes, the TDFA shows great promise for use in the implementation of CWDM systems.

REFERENCES

- [1] S. S.-H. Yam, Y. Akasaka, Y. Kubota, H. Inoue, K. Parameswaran. "Novel Pumping Schemes for Fluoride-Based Thulium-Doped Fiber Amplifier at 690 and 1050 nm (or 1400 nm)." *IEEE Photonics Technology Letters*, Vol 117, No 5, May 2005, pp 1001-1003.
- [2] T. Komukai, T. Yamamoto, T. Sugawa, Y. Miyajima. "Upconversion Pumped Thulium-Doped Fluoride Fiber Amplifier and Laser Operating at 1.47 μ m." *IEEE Journal of Quantum Electronics*, vol. 31, No 11. Nov. 1995.
- [3] T. Kasamatsu, Y. Yano, T. Ono. "1.49- μ m-Band Gain-Shifted Thulium-Doped Fiber Amplifier for WDM Transmission Systems." *Journal of Lightwave Technology*, vol 20, No 10, 2002, pp 1826-1838.
- [4] R.M. Percival, J.R. Williams. "Highly Efficient 1.064 μ m Upconversion Pumped Thulium-Doped Fiber Laser." *Electronics Letters*. Vol 30, Iss 19, 15 Sept 1994, pp 1598-1599.
- [5] F. Brunet, P. Laperle, R. Vallée, S. LaRochelle, L. Pujol. "Modeling of Tm-Doped ZBLAN Blue Upconversion Fiber Lasers Operating at 455 nm." *SPIE Proc.*, pp 125-135, 1999.
- [6] Y. Akasaka, H. Inoue, S. S.-H. Yam, Y. Kubota. "Thulium doped fiber amplifier for the first window from 790nm to 850nm with 690nm/1400nm dual pumping." *OFC Technical Digest*, vol. 3, March 2005, Paper OWF3.
- [7] A. S. L. Gomes et al. "Characterization of efficient dual-wavelength (1050 + 800 nm) pumping scheme for thulium-doped fiber amplifiers," *IEEE Photonics Technology Letters*, vol. 15, no. 2, pp. 200-202, Feb. 2003.
- [8] F. Roy, F. Leplindard, L. Lorcy, A. Le Sauze, P. Baniel, and D. Bayart. "48% power conversion efficiency in a singly-pump gain-shifted thulium doped fiber amplifier," in *OFC 2001*, Anaheim, CA, 2001, Paper PD2.
- [9] H. Inoue, K. Soga, A. Makishima. "Simulation of the Optical Properties of Tm:ZBLAN glass." *J. Non-Crystalline Solids*, vol. 336, Iss. 2, 2004, pp 135-147.
- [10] J. Sanz, R. Cases, R. Alcalá. "Optical Properties of Tm³⁺ in Fluorozirconate Glass." *J. Non-Crystalline Solids*, vol. 93, pp 377-386. 1987.
- [11] C. Guery, J.L. Adam, J. Lucas. "Optical Properties of Tm³⁺ in Indium-Based Fluoride Glasses." *J. Luminescence*, vol.42, pp 181-189. 1988.
- [12] Hewak, Dan. "Properties, Processing and Applications of Glass and Rare-Earth-Doped Glasses for Optical Fibres." *Institution of Engineering and Technology*, 1998, pp 170-179, 236-240.
- [13] S. S.-H. Yam, J. Kim. "Ground State Absorption in Thulium-Doped Fiber Amplifier: Experiment and Modeling." in press, 2006.
- [14] P.J. Corvini, et al. "Performance and reliability of high-power 670-690 nm CW laser diode bars grown by solid source molecular beam epitaxy," in *OSA Trends in Optics and Photonics, Advanced Semiconductor Lasers and Their Applications*, vol.31, 2000, pp 10-12.